

Complex Numbers

- **Definition 1**

The set of **complex numbers**, denoted by $\mathbb{C}$, is the set of all ordered pairs of real numbers with the sum and product defined by

$$\begin{aligned}(a, b) + (c, d) &= (a + c, b + d) \\ (a, b) \cdot (c, d) &= (ac - bd, ad + bc).\end{aligned}$$

That is, $\mathbb{C} = (\mathbb{R} \times \mathbb{R}, +, \cdot)$.

Since the function $\iota: \mathbb{R} \rightarrow \mathbb{C}$ defined by $\iota(a) = (a, 0)$ is a bijection, we can identify the real number a with the pair $(a, 0)$. Now, if we consider $1 = (1, 0)$ and $i = (0, 1)$, we have that

$$i^2 = (0, 1)(0, 1) = (-1, 0) = -1,$$

and every complex number can be written as

$$(a, b) = a(1, 0) + b(0, 1) = a + bi.$$

Hence, with this notation we have:

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)i \\ (a + bi) \cdot (c + di) &= (ac - bd) + (ad + bc)i\end{aligned}$$

and

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

- **Properties 1.1**

The arithmetic properties of $\mathbb{C}$ are as follows. Suppose $z, z_1, z_2, z_3 \in \mathbb{C}$:

1. $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3).$
2. $z + 0 = z.$
3. For each $z \in \mathbb{C}$, there exists a unique $-z \in \mathbb{C}$ such that $z + (-z) = 0.$
4. $z_1 + z_2 = z_2 + z_1.$
5. $(z_1 z_2) z_3 = z_1 (z_2 z_3).$
6. $z \cdot 1 = z.$
7. For each $z \in \mathbb{C}$ with $z \neq 0$, there exists a unique $z^{-1} \in \mathbb{C}$ such that $z \cdot z^{-1} = 1.$
8. $z_1 z_2 = z_2 z_1.$
9. $z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3.$

- **Definition 2**

Let $z = a + bi$. The real number a is called the **real part** of z and is denoted $\operatorname{Re}(z) = a$. The real number b is called the **imaginary part** of z and is denoted $\operatorname{Im}(z) = b$.

- **Definition 3**

Let $z = a + bi$. The **conjugate** of z , denoted $\bar{z}$, is defined by $\bar{z} = a - bi$.

The **modulus** of z , denoted $|z|$, is defined by $|z| = \sqrt{a^2 + b^2}$.

- **Remark 3.1**

$|z| \in \mathbb{R}$ and $|z| \geq 0$.

- **Theorem 3.2**

1. $|z|^2 = z\bar{z}$.
2. If $z \neq 0$, then $z^{-1} = \frac{\bar{z}}{|z|^2}$.
3. $\operatorname{Re}(z) = \frac{1}{2}(z + \bar{z})$.
4. $\operatorname{Im}(z) = \frac{1}{2i}(z - \bar{z})$.
5. $\overline{z + w} = \bar{z} + \bar{w}$.
6. $\overline{zw} = \bar{z}\bar{w}$.
7. $|\bar{z}| = |z|$.
8. $|zw| = |z||w|$.

■ **Proof 3.2.1**

1. If $z = a + bi$, then

$$z\bar{z} = (a + bi)(a - bi) = (a^2 + b^2) + (-ab + ab)i = a^2 + b^2 = |z|^2.$$

2. If $z \neq 0$, we have by Theorem 3.2(1):

$$z^{-1} = \frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2}.$$

3. If $z = a + bi$, then

$$\frac{1}{2}(z + \bar{z}) = \frac{1}{2}(a + bi + a - bi) = \frac{1}{2}(2a) = a = \operatorname{Re}(z).$$

4. If $z = a + bi$, then

$$\frac{1}{2i}(z - \bar{z}) = \frac{1}{2i}(a + bi - a + bi) = \frac{1}{2i}(2bi) = b = \operatorname{Im}(z).$$

5. If $z = a + bi, w = c + di$, then

$$\bar{z} + \bar{w} = (a - bi) + (c - di) = (a + c) - (b + d)i = \overline{(a + c) + (b + d)i} = \overline{z + w}.$$

6. If $z = a + bi, w = c + di$, then

$$\begin{aligned}\overline{zw} &= (a - bi)(c - di) \\ &= (ac - bd) + (-1)(-ad - bc)i \quad \text{since } i^2 = -1 \\ &= (ac - bd) - (ad + bc)i \\ &= \overline{(ac - bd) + (ad + bc)i} \\ &= \overline{zw}.\end{aligned}$$

7. If $z = a + bi$, then $\bar{z} = a - bi$. By the definition of modulus, we have:

$$|\bar{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|.$$

8. If $z, w \in \mathbb{C}$, then we have:

$$\begin{aligned}|zw|^2 &= zw \cdot \overline{zw} \\ &= zw \cdot \bar{z} \cdot \bar{w} \quad (\text{by Theorem 3.2(6)}) \\ &= \bar{z}\bar{z} \cdot w\bar{w} \\ &= |z|^2 |w|^2.\end{aligned}$$

Therefore, $|zw| = |z||w|$.

○ **Example 3.3**

If $z = a + bi$, find $\operatorname{Re}\left(\frac{1}{z}\right)$ and $\operatorname{Im}\left(\bar{z}^2 + z^2\right)$.

■ **Solution 3.3.1**

$$\text{Since } \frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{a - bi}{a^2 + b^2} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i, \text{ then } \operatorname{Re}\left(\frac{1}{z}\right) = \frac{a}{a^2 + b^2}.$$

Now, as

$$\begin{aligned}\bar{z}^2 + z^2 &= \bar{z}^2 + 2z\bar{z} + z^2 - 2z\bar{z} \\ &= (z + \bar{z})^2 - 2|z|^2 \\ &= (2\operatorname{Re}(z))^2 - 2|z|^2 \\ &= 4a^2 - 2(a^2 + b^2) \\ &= 2a^2 - 2b^2.\end{aligned}$$

Therefore, $\operatorname{Im}(\bar{z}^2 + z^2) = 0$.

◦ **Example 3.4**

If $z = a + bi$, express $\operatorname{Re}(iz)$ and $\operatorname{Im}((1+i)z)$ in terms of $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$.

■ **Solution 3.4.1**

Since $iz = i(a + bi) = -b + ai$, then $\operatorname{Re}(iz) = -b = -\operatorname{Im}(z)$.

Since $(1+i)z = (1+i)(a + bi) = (a-b) + (a+b)i$, then $\operatorname{Im}((1+i)z) = a+b = \operatorname{Re}(z) + \operatorname{Im}(z)$.

◦ **Proposition 3.5 (Triangle inequality on $\mathbb{C}$)**

If $z, w \in \mathbb{C}$, then

$$|z + w| \leq |z| + |w|.$$

■ **Proof 3.5.1**

$$\begin{aligned}|z + w|^2 &= (z + w)\overline{(z + w)} \\ &= (z + w)(\bar{z} + \bar{w}) \\ &= z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w} \\ &= |z|^2 + z\bar{w} + \bar{w}z + |w|^2 \\ &= |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \\ &\leq |z|^2 + 2|z\bar{w}| + |w|^2 \\ &= |z|^2 + 2|z||w| + |w|^2 \\ &= (|z| + |w|)^2.\end{aligned}$$

Therefore, $|z + w| \leq |z| + |w|$.

Given $z \in \mathbb{C}$ with $z \neq 0$, $z = a + bi = (a, b)$ has polar coordinates (r, θ) where

$$a = r \cos \theta, \quad b = r \sin \theta \quad (0 \leq \theta < 2\pi).$$

We have that:

$$\begin{aligned}\frac{b}{a} &= \frac{r \sin \theta}{r \cos \theta} = \tan \theta \implies \theta = \tan^{-1}\left(\frac{b}{a}\right). \\ a^2 + b^2 &= r^2. \\ r &= |z| \geq 0. \\ \cos \theta &= \frac{a}{\sqrt{a^2 + b^2}}, \quad \sin \theta = \frac{b}{\sqrt{a^2 + b^2}}.\end{aligned}$$

Note that:

$$\begin{aligned}z &= |z| \cdot \frac{z}{|z|} \\ &= r \left(\frac{a + bi}{\sqrt{a^2 + b^2}} \right) \\ &= r \left(\frac{a}{\sqrt{a^2 + b^2}} + i \frac{b}{\sqrt{a^2 + b^2}} \right) \\ &= r(\cos \theta + i \sin \theta).\end{aligned}$$

- **Definition 4**

If $z \in \mathbb{C}$ with $z \neq 0$, the expression

$$z = r(\cos \theta + i \sin \theta) \quad (\theta \in \mathbb{R})$$

is called the **polar representation** of z .

- **Notation 4.1**

$$\text{cis } \theta = \cos \theta + i \sin \theta.$$

- **Definition 5**

A real number θ that satisfies $z = r \text{cis } \theta$ is called an **argument** of z . The unique argument $\theta \in [0, 2\pi)$ is called the **principal argument** of z and is denoted $\text{Arg}(z)$.

- **Remark 5.1**

Since $\cos(\theta + 2k\pi) = \cos \theta$ and $\sin(\theta + 2k\pi) = \sin \theta$ for all $k \in \mathbb{Z}$, we have that every argument of z has the form

$$\text{Arg}(z) + 2k\pi.$$

The argument of 0 is undefined.

- **Example 5.2**

If $z = i$, then $\text{Arg}(z) = \frac{\pi}{2}$ and

$$z = |i| \left(\cos \left(\frac{\pi}{2} \right) + i \sin \left(\frac{\pi}{2} \right) \right).$$

- **Theorem 5.3**

Let $z, w \in \mathbb{C}$ with $z, w \neq 0$ with polar representations

$$z = r \text{cis } \theta, \quad w = s \text{cis } \varphi.$$

Then zw has the polar representation

$$zw = rs \text{cis } (\theta + \varphi).$$

And $\frac{z}{w}$ has the polar representation

$$\frac{z}{w} = \frac{r}{s} \text{cis } (\theta - \varphi).$$

- **Proof 5.3.1**

For the product zw , by Definition 4 and the trigonometric addition formulas, we have:

$$\begin{aligned} zw &= (r \text{cis } \theta)(s \text{cis } \varphi) \\ &= rs(\cos \theta + i \sin \theta)(\cos \varphi + i \sin \varphi) \\ &= rs((\cos \theta \cos \varphi - \sin \theta \sin \varphi) + i(\sin \theta \cos \varphi + \cos \theta \sin \varphi)) \\ &= rs(\cos(\theta + \varphi) + i \sin(\theta + \varphi)) \\ &= rs \text{cis}(\theta + \varphi). \end{aligned}$$

For the quotient $\frac{z}{w}$, since $w \cdot w^{-1} = 1$, we first find the polar representation of $w^{-1} = \frac{\overline{w}}{|w|^2}$:

$$w^{-1} = \frac{s(\cos \varphi - i \sin \varphi)}{s^2} = \frac{1}{s}(\cos(-\varphi) + i \sin(-\varphi)) = \frac{1}{s} \text{cis}(-\varphi).$$

Applying the product rule proved above, we obtain:

$$\frac{z}{w} = z \cdot w^{-1} = (r \text{cis } \theta) \cdot \left(\frac{1}{s} \text{cis}(-\varphi) \right) = \frac{r}{s} \text{cis}(\theta - \varphi).$$

- **Corollary 5.4**

If $z_1, z_2, \dots, z_n \in \mathbb{C}$ with $z_1, z_2, \dots, z_n \neq 0$ and polar representations

$$z_1 = r_1 \operatorname{cis} \theta_1, \quad z_2 = r_2 \operatorname{cis} \theta_2, \dots, \quad z_n = r_n \operatorname{cis} \theta_n.$$

Then $z_1 z_2 \cdots z_n$ has the polar representation

$$z_1 z_2 \cdots z_n = r_1 r_2 \cdots r_n \operatorname{cis} (\theta_1 + \theta_2 + \cdots + \theta_n).$$

■ **Proof 5.4.1**

By induction on n .

■ **Base step:** For $n = 1$, the statement is trivially true since $z_1 = r_1 \operatorname{cis} \theta_1$.

■ **Inductive step:** Suppose that the result is true for n , that is, $z_1 z_2 \cdots z_n = r_1 r_2 \cdots r_n \operatorname{cis} (\theta_1 + \theta_2 + \cdots + \theta_n)$. For $n + 1$, by Theorem 5.3, we have:

$$\begin{aligned} (z_1 z_2 \cdots z_n) \cdot z_{n+1} &= (r_1 r_2 \cdots r_n \operatorname{cis} (\theta_1 + \cdots + \theta_n)) \cdot (r_{n+1} \operatorname{cis} \theta_{n+1}) \\ &= (r_1 r_2 \cdots r_n \cdot r_{n+1}) \operatorname{cis} ((\theta_1 + \cdots + \theta_n) + \theta_{n+1}) \\ &= r_1 r_2 \cdots r_{n+1} \operatorname{cis} (\theta_1 + \theta_2 + \cdots + \theta_{n+1}). \end{aligned}$$

Thus, the result is true for $n + 1$. So the result is true for every $n \in \mathbb{N}$.

○ **Corollary 5.5 (De Moivre's formula)**

If $z \in \mathbb{C}$ with $z \neq 0$ and polar representation $z = r \operatorname{cis} \theta$, then z^n has the polar representation

$$z^n = r^n \operatorname{cis}(n\theta) \quad (n \in \mathbb{Z}).$$

■ **Proof 5.5.1**

We first prove the case for $n \in \mathbb{N}_0$ by induction on n .

■ **Base step:** If $n = 0$, then $z^0 = 1$ and $r^0 \operatorname{cis}(0 \cdot \theta) = 1 \cdot (\cos 0 + i \sin 0) = 1$. The result is true.

■ **Inductive step:** Suppose that the result is true for $n \in \mathbb{N}_0$, that is, $z^n = r^n \operatorname{cis}(n\theta)$. For $n + 1$, by Definition 3 and Theorem 5.3, we have:

$$z^{n+1} = z^n \cdot z = (r^n \operatorname{cis}(n\theta)) \cdot (r \operatorname{cis} \theta) = (r^n \cdot r) \operatorname{cis}(n\theta + \theta) = r^{n+1} \operatorname{cis}((n+1)\theta).$$

Thus, the formula holds for all $n \in \mathbb{N}_0$.

Now, consider the case where n is a negative integer, so $n = -m$ for some $m \in \mathbb{N}$. By Definition 3 and Theorem 5.3, we have:

$$z^n = z^{-m} = (z^{-1})^m = \left(\frac{1}{r} \operatorname{cis}(-\theta) \right)^m.$$

Since $m \in \mathbb{N}$, applying the formula proved for natural numbers yields:

$$\left(\frac{1}{r} \operatorname{cis}(-\theta) \right)^m = \left(\frac{1}{r} \right)^m \operatorname{cis}(m(-\theta)) = r^{-m} \operatorname{cis}(-m\theta) = r^n \operatorname{cis}(n\theta).$$

Therefore, De Moivre's formula holds for all $n \in \mathbb{Z}$.

○ **Example 5.6**

Since $\frac{\sqrt{3}}{2} + \frac{1}{2}i = \operatorname{cis} \left(\frac{\pi}{6} \right)$, then by De Moivre's formula we have that:

$$\begin{aligned} \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)^3 &= \operatorname{cis} \left(3 \cdot \frac{\pi}{6} \right) \\ &= \operatorname{cis} \left(\frac{\pi}{2} \right) \\ &= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \\ &= i. \end{aligned}$$